

Elementary Problems

Sequence

Solutions

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Basic Work

1. The common difference d equals

$$\frac{200 - 20}{12 - 3} = 20.$$

Thus, $a_1 = a_3 - 2d = -20$.

2. We have $a_n = a_1 + 3(n - 1) = 3n - 2$. Thus,

$$S_n = 1 + 4 + 7 + \cdots + (3n - 2) = \frac{(1 + (3n - 2))n}{2} = \frac{(3n - 1)n}{2}.$$

3. (a) We guess $1^2 + 2^2 + \cdots + n^2 = P(n)$ for some polynomial $P(n)$. If we want to prove this by induction, then we need to show $P(n) + (n + 1)^2 = P(n + 1)$ in the inductive step. If $P(n)$ is of degree m , then $P(n + 1) - P(n)$ is of degree $m - 1$. Thus, we have $m = 3$.

Let $P(n) = an^3 + bn^2 + cn + d$. As $P(0) = 0$, we have $d = 0$. The above identity becomes

$$an^3 + bn^2 + cn + (n + 1)^2 = a(n + 1)^3 + b(n + 1)^2 + c(n + 1).$$

Comparing the coefficients, we need

$$\begin{aligned} a &= a, \\ b + 1 &= 3a + b, \\ c + 2 &= 3a + 2b + c, \\ 1 &= a + b + c. \end{aligned}$$

One easily checks that the only solution is $a = \frac{1}{3}$, $b = \frac{1}{2}$ and $c = \frac{1}{6}$. Thus,

$$1^2 + 2^2 + \cdots + n^2 = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n = \frac{n(n + 1)(2n + 1)}{6}.$$

The formula can be proved via induction with the base case $n = 0$ from above.

- (b) Instead of the inductive approach, we present another proof here. Note that

$$(k + 1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1$$

for $k = 1, 2, \dots, n$. Summing over all these k 's, we have

$$\begin{aligned} & (n + 1)^4 - 1 \\ &= (2^4 - 1^4) + (3^4 - 2^4) + \cdots + ((n + 1)^4 - n^4) \\ &= 4(1^3 + 2^3 + \cdots + n^3) + 6(1^2 + 2^2 + \cdots + n^2) + 4(1 + 2 + \cdots + n) + n. \end{aligned}$$

Therefore,

$$\begin{aligned}
& 1^3 + 2^3 + \cdots + n^3 \\
&= \frac{1}{4} ((n+1)^4 - 1 - 6(1^2 + 2^2 + \cdots + n^2) - 4(1 + 2 + \cdots + n) - n) \\
&= \frac{1}{4} ((n+1)^4 - 1 - n(n+1)(2n+1) - 2n(n+1) - n) \\
&= \frac{n^4 + 2n^3 + n^2}{4} = \frac{n^2(n+1)^2}{4}.
\end{aligned}$$

(c) We have

$$\begin{aligned}
& 1 \times 2 + 2 \times 3 + \cdots + n(n+1) \\
&= \sum_{k=1}^n k(k+1) = \sum_{k=1}^n k^2 + \sum_{k=1}^n k \\
&= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{3}.
\end{aligned}$$

4. Note that $\frac{a_8}{a_6} = \frac{a_6}{a_4} = \frac{a_4}{a_2} = 2$. Therefore,

$$a_8 = \frac{a_8}{a_6} \times \frac{a_6}{a_4} \times \frac{a_4}{a_2} \times a_2 = 2 \times 2 \times 2 \times 2 = 16.$$

5. We have

$$S_n = a_1 + a_2 + \cdots + a_n.$$

Multiplying both sides by 3, as $3a_k = a_{k+1}$, we have

$$3S_n = a_2 + a_3 + \cdots + a_{n+1}.$$

Thus,

$$2S_n = (a_2 + a_3 + \cdots + a_{n+1}) - (a_1 + a_2 + \cdots + a_n) = a_{n+1} - a_1 = 3^n - 1.$$

It follows that $S_n = \frac{3^n - 1}{2}$.

6. Let

$$S = a_1 + a_2 + \cdots.$$

Multiplying both sides by $\frac{1}{3}$, as $\frac{1}{3}a_k = a_{k+1}$, we have

$$\frac{1}{3}S = a_2 + a_3 + \cdots.$$

Thus,

$$\frac{2}{3}S = (a_1 + a_2 + \cdots) - (a_2 + a_3 + \cdots) = a_1 = 1.$$

It follows that $S = \frac{3}{2}$.

7. Let the common difference be d . Then the first three terms are $a_1, a_1 + d, a_1 + 2d$. As it is a geometric sequence, we have

$$\frac{a_1 + d}{a_1} = \frac{a_1 + 2d}{a_1 + d}.$$

After expansion, this becomes $d^2 = 0$. Thus, we must have $d = 0$. This means the sequence is a constant sequence. Conversely, such a sequence is an arithmetic sequence as well as a geometric sequence (the zero sequence may or may not be considered as a geometric sequence).

8. Let $a = c - 2x$, $b = c - x$, $d = c + y$ and $e = c + 2y$ where $x, y \geq 0$. Since $ae = c^2$ and $bd = c^2$, we have

$$(c - 2x)(c + 2y) = c^2 \quad \text{and} \quad (c - x)(c + y) = c^2.$$

These can be simplified to $cy = cx + 2xy$ and $cy = cx + xy$. Comparing these, we get $xy = 0$. Thus, $x = 0$ or $y = 0$. From $cy = cx + xy$, both cases imply $x = y = 0$ since $c > 0$. This means $a = b = c = d = e$.

Conversely, when $a = b = c = d = e > 0$, the conditions are clearly satisfied.

9. We have

$$\begin{aligned} & a_1 + a_2 + \cdots \\ &= \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \cdots \\ &= \frac{1}{2} + \left(\frac{1}{2^2} + \frac{1}{2^2} \right) + \left(\frac{1}{2^3} + \frac{1}{2^3} + \frac{1}{2^3} \right) + \cdots \\ &= \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots \right) + \left(\frac{1}{2^2} + \frac{1}{2^3} + \cdots \right) + \left(\frac{1}{2^3} + \cdots \right) + \cdots \\ &= 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots = 2. \end{aligned}$$

Note that each bracket in the second last line can be computed using the sum of geometric sequences.

An equivalent way to write is that

$$\sum_{k=1}^{\infty} \frac{k}{2^k} = \sum_{k=1}^{\infty} \sum_{m=1}^k \frac{1}{2^k} = \sum_{m=1}^{\infty} \sum_{k=m}^{\infty} \frac{1}{2^k} = \sum_{m=1}^{\infty} 2^{1-m} = 2.$$

10. Note that

$$a_n = \frac{1}{(3n-2)(3n+1)} = \frac{1}{3} \left(\frac{1}{3n-2} - \frac{1}{3n+1} \right).$$

Thus,

$$S_n = \frac{1}{3} \left(\frac{1}{1} - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} + \cdots + \frac{1}{3n-2} - \frac{1}{3n+1} \right) = \frac{1}{3} \left(1 - \frac{1}{3n+1} \right) = \frac{n}{3n+1}.$$

11. Let

$$a_n = \frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+2}$$

for some constants A, B, C . This is equivalent to

$$1 = A(n+1)(n+2) + Bn(n+2) + Cn(n+1).$$

Putting $n = 0$, we get $1 = A(1)(2)$, i.e. $A = \frac{1}{2}$. Similarly, by putting $n = -1$ and $n = -2$ respectively, we get $B = -1$ and $C = \frac{1}{2}$. Thus, for $n \geq 2$,

$$\begin{aligned} S_n &= \frac{1}{2} \left(\frac{1}{1} - 2 \cdot \frac{1}{2} + \frac{1}{3} + \frac{1}{2} - 2 \cdot \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n} - 2 \cdot \frac{1}{n+1} + \frac{1}{n+2} \right) \\ &= \frac{1}{2} \left(1 - \frac{1}{2} - \frac{1}{n+1} + \frac{1}{n+2} \right) = \frac{n^2 + 3n}{4(n+1)(n+2)}. \end{aligned}$$

Note that this also holds when $n = 1$.

12. From $n^4 + n^2 + 1 = (n^2 - n + 1)(n^2 + n + 1)$, we may consider

$$a_n = \frac{A}{n^2 - n + 1} + \frac{B}{n^2 + n + 1}.$$

One easily finds that $A = \frac{1}{2}$ and $B = -\frac{1}{2}$. Note that $n^2 - n + 1 = (n-1)n + 1$ and $n^2 + n + 1 = n(n+1) + 1$. Thus,

$$\begin{aligned} S_n &= \frac{1}{2} \left(\frac{1}{0 \cdot 1 + 1} - \frac{1}{1 \cdot 2 + 1} + \frac{1}{1 \cdot 2 + 1} - \frac{1}{2 \cdot 3 + 1} + \cdots \right. \\ &\quad \left. + \frac{1}{(n-1)n + 1} - \frac{1}{n(n+1) + 1} \right) \\ &= \frac{1}{2} \left(1 - \frac{1}{n^2 + n + 1} \right) = \frac{n^2 + n}{2(n^2 + n + 1)}. \end{aligned}$$

13. Firstly, if $a_{n+1} - a_n = d > 0$ for some $n \geq 1$, then we have $a_{n+2} \geq 2a_{n+1} - a_n = a_{n+1} + d$. Inductively, we can show that $a_{k+1} \geq a_k + d$ for $k \geq n$. Thus,

$$a_m = (a_m - a_{m-1}) + (a_{m-1} - a_{m-2}) + \cdots + (a_{n+1} - a_n) + a_n \geq d(m-n) + a_n.$$

When m tends to infinity, this lower bound tends to infinity, which contradicts the fact that $a_m \leq 1$. Hence, we must have $(n+1)(a_n - a_{n+1}) \geq 0$.

Secondly, suppose $a_n - a_{n+1} > \frac{2}{n+1}$ for some $n \geq 1$. Then we have

$$a_{n-1} - a_n \geq a_n - a_{n+1} > \frac{2}{n+1}.$$

Inductively, we find that $a_k - a_{k+1} > \frac{2}{n+1}$ for $k = n, n-1, \dots, 1$. Adding these inequalities, we obtain

$$a_1 - a_{n+1} = (a_1 - a_2) + (a_2 - a_3) + \dots + (a_n - a_{n+1}) > \frac{2n}{n+1}.$$

As $0 \leq a_1, a_{n+1} \leq 1$, this implies $1 > \frac{2n}{n+1}$. This is impossible. Thus,

$$(n+1)(a_n - a_{n+1}) \leq 2$$

for any $n \geq 1$.

14. (a) False. For example, the constant sequence defined by $a_n = 1$ for any $n \geq 1$ is an arithmetic sequence which is bounded. However, an arithmetic sequence with nonzero common difference is unbounded.
- (b) (Real numbers) False. For example, the sequence defined by $a_n = 2^{-n}$ for any $n \geq 1$ is a geometric sequence which is bounded as $0 < a_n < 1$ for any $n \geq 1$. However, an arithmetic sequence with common ratio r such that $|r| > 1$ is unbounded.
- (Positive integers) True. For a non-constant geometric sequence of positive integers, the common ratio r must satisfy $r > 1$. Indeed, suppose $r < 1$ and the leading term is $a > 0$. Then the general term is ar^{n-1} , which is positive but less than 1 for sufficiently large n . This contradicts the fact that it is a positive integer. For $r > 1$, the general term is ar^{n-1} , which is unbounded.
- (c) False. For example, the sequence defined by $a_n = n$ for any $n \geq 1$ is a monotonic sequence which is unbounded.
- (d) True. By definition of a periodic sequence, there exists $d > 0$ such that $a_{n+d} = a_n$ for any $n \geq 1$. Therefore, for each $n \geq 1$, we have $a_n = a_r$ where $n \equiv r \pmod{d}$. Thus, each a_n is equal to one of

$$a_1, a_2, \dots, a_d.$$

In particular, the absolute value of each term is bounded by

$$M = \max \{|a_1|, |a_2|, \dots, |a_d|\}.$$

- (e) False. For example, the sequence defined by $a_1 = a_2 = 1$ and $a_{n+2} = 2a_{n+1} - a_n$ for $n \geq 1$ is bounded since one can prove by induction that $a_n = 1$ for any $n \geq 1$.
- (f) False. Consider the decimal representation of $\sqrt{2}$, i.e. $1.414213\dots$. We define $a_1 = 1$ and define a_n to be the $(n-1)$ st decimal place of $\sqrt{2}$ for $n \geq 2$. The sequence is bounded since $0 \leq a_n \leq 9$ for each n .
- Suppose on the contrary that this sequence is periodic. Then there exists $d > 0$ such that $a_{n+d} = a_n$ for any $n \geq 1$. From

$$\sqrt{2} = \overline{a_1.a_2a_3a_4\dots},$$

we have

$$10^d \sqrt{2} = \overline{a_1 a_2 \cdots a_{d+1} . a_{d+2} a_{d+3} a_{d+4} \cdots}.$$

Taking their differences, we find that

$$(10^d - 1)\sqrt{2} = \overline{a_1 a_2 \cdots a_{d+1}} - a_1$$

since all decimal places are identical. It follows from $d > 0$ that $\sqrt{2}$ is a rational number. This is a contradiction. (For a counterexample of positive integers, one may simply consider the sequence $(b_n)_{n \geq 1}$ defined by $b_n = a_n + 1$ for any $n \geq 1$.)

- (g) (Real numbers) False. For example, the sequence defined by $a_n = 2^{-n}$ for any $n \geq 1$ is bounded since $0 < a_n < 1$ for any $n \geq 1$. However, it does not contain equal terms since it is obvious that

$$a_1 > a_2 > \cdots .$$

(Positive integers) True. Suppose $a_n \leq M$ for any $n \geq 1$ where M is a positive constant. Since each a_n is a positive integer, each of them can only be one of

$$1, 2, \dots, [M].$$

As there are infinitely many terms, there must be infinitely many equal terms by the pigeonhole principle.

- (h) (Real numbers) False. For example, the sequence defined by $a_n = 1 - 2^{-n}$ for any $n \geq 1$ is bounded as $0 < a_n < 1$ for any $n \geq 1$. However, it does not have a largest term since it is obvious that

$$a_1 < a_2 < \cdots .$$

(Positive integers) True. Suppose $a_n \leq M$ for any $n \geq 1$ where M is a positive constant. Since each a_n is a positive integer, each of them can only be one of

$$1, 2, \dots, [M].$$

There is a largest term since any finite set has a maximum. Also, any sequence (bounded or not) of positive integers must have a smallest term as this is a fundamental property of the set of positive integers.

- (i) True. Suppose a_M is one of the largest terms. This means $a_n \leq a_M$ for any $n \geq 1$. Thus, a_M is an upper bound of the sequence.
- (j) (Real numbers) False. For example, the sequence defined by $a_n = \frac{1}{n}$ for any $n \geq 1$

satisfies $\lim_{n \rightarrow \infty} a_n = 0$. However, the partial sums are unbounded. Indeed, we have

$$\begin{aligned}
& a_1 + a_2 + \cdots + a_{2^k} \\
&= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots \\
&\quad + \left(\frac{1}{2^{k-1}+1} + \frac{1}{2^{k-1}+2} + \cdots + \frac{1}{2^k}\right) \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^k} + \frac{1}{2^k} + \cdots + \frac{1}{2^k}\right) \\
&= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2} = 1 + \frac{k}{2}.
\end{aligned}$$

When k tends to ∞ , the partial sums are arbitrarily large.

(Positive integers) True. The terms of a sequence of positive integers are at least 1. Thus, they cannot be arbitrarily close to 0. By the law of logic, the hypothesis is false and so the statement is true.

(k) (Real numbers) False. For example, the sequence defined by $a_n = (-1)^n$ for any $n \geq 1$ has bounded partial sums since one can show that $a_1 + a_2 + \cdots + a_n$ is -1 if n is odd and is 0 if n is even. However, the terms are not arbitrarily close to 0 .

(Positive integers) True. The partial sums of a sequence of positive integers cannot be bounded since $a_1 + a_2 + \cdots + a_n \geq n$ for any $n \geq 1$. By the law of logic, the hypothesis is false and so the statement is true.

Recurrence Sequence

15. We have

$$\begin{aligned}
a_n &= 2a_{n-1} + 2^{n-1}, \\
2a_{n-1} &= 2^2a_{n-2} + 2^{n-1}, \\
2^2a_{n-2} &= 2^3a_{n-3} + 2^{n-1}, \\
&\dots, \\
2^{n-1}a_1 &= 2^na_0 + 2^{n-1}.
\end{aligned}$$

Adding these equations, we obtain $a_n = 2^na_0 + n \cdot 2^{n-1} = n \cdot 2^{n-1}$. If $n = 2^k$ for some integer $k \geq 0$, then $a_n = 2^{k+n-1}$ is a power of 2.

16. We have

$$\begin{aligned}
a_n &= 2a_{n-1} + (n-1)(1 + 2^{n-1}), \\
2a_{n-1} &= 2^2a_{n-2} + 2(n-2)(1 + 2^{n-2}), \\
2^2a_{n-2} &= 2^3a_{n-3} + 2^2(n-3)(1 + 2^{n-3}), \\
&\dots, \\
2^{n-2}a_2 &= 2^{n-1}a_1 + 2^{n-2}(1)(1 + 2^1).
\end{aligned}$$

Adding these equations, we obtain

$$\begin{aligned}
a_n &= 2^{n-1}a_1 + (n-1)(1+2^{n-1}) + (n-2)(2+2^{n-1}) + \cdots + (1)(2^{n-2}+2^{n-1}) \\
&= 2^{n-1} + \sum_{k=1}^{n-1} k \cdot 2^{n-1-k} + \sum_{k=1}^{n-1} k \cdot 2^{n-1} \\
&= 2^{n-1} + \sum_{k=1}^{n-1} \sum_{m=1}^k 2^{n-1-k} + 2^{n-1} \cdot \frac{(n-1)n}{2} \\
&= 2^{n-1} + \sum_{m=1}^{n-1} \sum_{k=m}^{n-1} 2^{n-1-k} + 2^{n-2}(n-1)n \\
&= 2^{n-1} + \sum_{m=1}^{n-1} (2^{n-m} - 1) + 2^{n-2}(n-1)n \\
&= 2^{n-1} + 2^n - 2 - (n-1) + 2^{n-2}(n-1)n \\
&= 2^{n-2}(n^2 - n + 6) - n - 1.
\end{aligned}$$

17. We claim that $a_n = 2 \cdot 3^{n-2}$ for $n \geq 2$. The base case $n = 2$ can be verified directly. Assume $a_j = 2 \cdot 3^{j-2}$ for all $j = 2, 3, \dots, k$ where $k \geq 2$. Then

$$\begin{aligned}
a_{k+1} &= 2(a_1 + a_2 + \cdots + a_k) = 2(1 + 2 + 2 \cdot 3 + 2 \cdot 3^2 + \cdots + 2 \cdot 3^{k-2}) \\
&= 2 \left(1 + \frac{2(3^{k-1} - 1)}{3 - 1} \right) = 2 \cdot 3^{k-1}.
\end{aligned}$$

This establishes the case $n = k + 1$. By induction, $a_n = 2 \cdot 3^{n-2}$ for any $n \geq 2$.

18. We claim that $a_n = 3^n + 1$ for $n \geq 0$. The base cases $n = 0, 1$ hold trivially. Suppose $a_j = 3^j + 1$ for any $j \leq k$ where $k \geq 1$. For the case $n = k + 1$, we have

$$a_{k+1} = 3a_k - 2a_{k-1} + 2 \times 3^{k-1} = 3(3^k + 1) - 2(3^{k-1} + 1) + 2 \times 3^{k-1} = 3^{k+1} + 1.$$

This establishes the case $n = k + 1$. By induction, $a_n = 3^n + 1$ for any $n \geq 0$.

19. We claim that $a_n = \frac{6n-1}{3n-2}$ for $n \geq 1$. The base case $n = 1$ holds trivially. Suppose $a_k = \frac{6k-1}{3k-2}$ for some $k \geq 1$. Then

$$a_{k+1} = \frac{3a_k - 4}{a_k - 1} = \frac{3(6k-1) - 4(3k-2)}{(6k-1) - (3k-2)} = \frac{6k+5}{3k+1}.$$

This establishes the case $n = k + 1$. By induction, $a_n = \frac{6n-1}{3n-2}$ for any $n \geq 1$.

20. We define $n!! = n(n-2)(n-4)\cdots$ for each positive integer n . The product ends at 1 if n is odd and ends at 2 if n is even. By considering small cases, we claim that $a_n = \frac{(n-1)!!}{(n-2)!!}$ for $n \geq 3$. The base cases $n = 3$ and $n = 4$ can be verified directly. Assume this holds for $n = k$ and $n = k+1$. Then

$$a_{k+2} = \frac{1}{a_{k+1}} + a_k = \frac{(k-1)!!}{k!!} + \frac{(k-1)!!}{(k-2)!!} = \frac{(k-1)!!}{k!!}(1+k) = \frac{(k+1)!!}{k!!}.$$

By induction, the formula holds for any $n \geq 3$. Thus, $a_{2004} = \frac{2003!!}{2002!!}$.

21. We prove that

$$\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} = 1 - \frac{1}{a_1 a_2 \cdots a_n} \quad (1)$$

for each $n \geq 1$. This clearly holds when $n = 1$. Assume (1) holds when $n = k$ for some $k \geq 1$. We have

$$\begin{aligned} \frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} + \frac{1}{a_{k+1}} &= 1 - \frac{1}{a_1 a_2 \cdots a_k} + \frac{1}{a_{k+1}} \\ &= 1 - \frac{1}{a_1 a_2 \cdots a_k} + \frac{1}{a_1 a_2 \cdots a_k + 1} \\ &= 1 - \frac{1}{a_1 a_2 \cdots a_k (a_1 a_2 \cdots a_k + 1)} \\ &= 1 - \frac{1}{a_1 a_2 \cdots a_{k+1}}. \end{aligned}$$

By induction, (1) holds for any $n \geq 1$. Thus,

$$a_1 a_2 \cdots a_n \left(1 - \frac{1}{a_1} - \frac{1}{a_2} - \cdots - \frac{1}{a_n} \right) = 1.$$

22. The relation can be rewritten as $1 - a_{n+1} = a_n(1 - a_n)$. Thus,

$$\begin{aligned} 1 - a_{n+1} &= a_n(1 - a_n) = a_n a_{n-1}(1 - a_{n-1}) = \cdots = a_n a_{n-1} \cdots a_1(1 - a_1) \\ &= a_n a_{n-1} \cdots a_0. \end{aligned}$$

Now, we prove that

$$\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_n} = \frac{1}{a_0 a_1 \cdots a_n} \quad (1)$$

for each $n \geq 0$. This clearly holds when $n = 0$. Assume (1) holds when $n = k$ for some

$k \geq 0$. We have

$$\begin{aligned} \frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_k} + \frac{1}{a_{k+1}} &= \frac{1}{a_0 a_1 \cdots a_k} + \frac{1}{a_{k+1}} \\ &= \frac{1}{a_0 a_1 \cdots a_k} + \frac{1}{1 - a_0 a_1 \cdots a_k} \\ &= \frac{1}{a_0 a_1 \cdots a_k (1 - a_0 a_1 \cdots a_k)} \\ &= \frac{1}{a_0 a_1 \cdots a_{k+1}}. \end{aligned}$$

By induction, (1) holds for any $n \geq 0$. Thus,

$$a_0 a_1 \cdots a_n \left(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_n} \right) = 1.$$

23. We claim that $x_n \leq \frac{n}{2}$ for any $n \geq 1$. The base case $n = 1$ is trivial. Assume $x_k \leq \frac{k}{2}$ for some $k \geq 1$. Then

$$x_{k+1} = x_k + \frac{x_k^2}{k^2} \leq \frac{k}{2} + \frac{1}{k^2} \cdot \frac{k^2}{4} = \frac{k}{2} + \frac{1}{4} < \frac{k+1}{2}.$$

By induction, $x_n \leq \frac{n}{2}$ for any $n \geq 1$. Thus, $x_{2001} \leq \frac{2001}{2} < 1001$.

24. We claim that

$$a_1 + a_2 + \cdots + a_n = 1 - (2n - 1)a_n \quad (1)$$

for any $n \geq 1$. The base case $n = 1$ is trivial. Assume (1) holds when $n = k$ where $k \geq 1$. Then

$$\begin{aligned} a_1 + a_2 + \cdots + a_k + a_{k+1} &= 1 - (2k - 1)a_k + a_{k+1} \\ &= 1 - (2k + 2)a_{k+1} + a_{k+1} \\ &= 1 - (2k + 1)a_{k+1}. \end{aligned}$$

This establishes the case $n = k + 1$. By induction, (1) holds for any $n \geq 1$. Thus,

$$a_1 + a_2 + \cdots + a_n < 1.$$

25. By observing the pattern, we claim that

$$f_n = (n - 1)c + (n - 2)^2 \quad (1)$$

for any $n \in \mathbb{N}$. The base cases $n = 1$ and $n = 2$ are trivial. Assume this holds when $n = k, k + 1$ for some $k \geq 1$. Then

$$\begin{aligned} f_{k+2} &= 2f_{k+1} - f_k + 2 \\ &= 2(kc + (k - 1)^2) - ((k - 1)c + (k - 2)^2) + 2 \\ &= (k + 1)c + k^2. \end{aligned}$$

This establishes the case $n = k + 2$. By induction, (1) holds for any $n \in \mathbb{N}$.

Firstly, we have

$$\begin{aligned} f_k f_{k+1} &= ((k-1)c + (k-2)^2)(kc + (k-1)^2) \\ &= k(k-1)c^2 + (2k^3 - 7k^2 + 7k - 1)c + (k-1)^2(k-2)^2. \end{aligned} \quad (2)$$

Secondly, we consider $r = ac + b$ for some a, b in terms of k . Then

$$f_r = (r-1)c + (r-2)^2 = (a^2 + a)c^2 + (2ab - 4a + b - 1)c + (b-2)^2. \quad (3)$$

Comparing (2) and (3), we can take $a = k-1$ and $b = k^2 - 3k + 4$ so that all coefficients on the right-hand sides are equal. In other words, we have

$$f_k f_{k+1} = f_{(k-1)c + (k^2 - 3k + 4)}.$$

26. (1) No. Suppose such a sequence exists. The condition can be rewritten as

$$\frac{1}{2} \cdot \frac{a_{n+1}}{a_n} \geq \frac{a_{n+2}}{a_{n+1}}.$$

Thus, we have

$$\frac{a_n}{a_{n-1}} \leq \frac{1}{2} \cdot \frac{a_{n-1}}{a_{n-2}} \leq \frac{1}{2^2} \cdot \frac{a_{n-2}}{a_{n-3}} \leq \dots \leq \frac{1}{2^{n-2}} \cdot \frac{a_2}{a_1}.$$

Since a_1, a_2 are fixed, the right-hand side is less than 1 for all sufficiently large n . Hence, $a_n < a_{n-1}$ for all sufficiently large n . However, it is impossible to have a strictly decreasing sequence of positive integers. This is a contradiction.

(2) Yes. For example, we can define $a_1 = a_2 = 1$ and $a_{n+2} = \frac{a_{n+1}^2}{4a_n}$ for $n \geq 1$. Then we have

$$2a_n a_{n+2} = \frac{a_{n+1}^2}{2} \leq a_{n+1}^2$$

for any positive integer n . Also, it is obvious that a_n is rational by induction.

27. If $a_1 \geq 2$, we claim that $a_n \geq a_{n+1} \geq 2$ for any $n \geq 1$. Firstly, we have

$$a_2 = \sqrt{a_1 + 2} \geq \sqrt{2 + 2} = 2.$$

Also, it is obvious that $(a_1 + 1)(a_1 - 2) \geq 0$. This can be rewritten as $a_1^2 \geq a_1 + 2 = a_2^2$. Since $a_1, a_2 > 0$, this implies $a_1 \geq a_2$. These establish the base case $a_1 \geq a_2 \geq 2$.

Assume $a_k \geq a_{k+1} \geq 2$ for some $k \geq 1$. Using the same method, we find that

$$a_{k+2} = \sqrt{a_{k+1} + 2} \geq \sqrt{2 + 2} = 2.$$

Also, from $(a_{k+1} + 1)(a_{k+1} - 2) \geq 0$, we deduce $a_{k+1}^2 \geq a_{k+2}^2$ and hence $a_{k+1} \geq a_{k+2}$. These establish the inductive step $a_{k+1} \geq a_{k+2} \geq 2$.

By induction, we prove that $a_n \geq a_{n+1} \geq 2$ for any $n \geq 1$. In particular, this implies $a_1 \geq a_2 \geq \dots$.

Similarly, if $a_1 \leq 2$, the above proof goes the same by replacing $\geq$ with $\leq$. Thus, we obtain $a_1 \leq a_2 \leq \dots$ in this case.

In any case, the sequence (a_n) is monotonic.

28. We shall prove that $F_{n+1}^2 - F_n F_{n+2} = -(F_n^2 - F_{n-1} F_{n+1})$ for $n \geq 1$. Rearranging the terms, we need to prove

$$F_{n+1}(F_{n+1} - F_{n-1}) = F_n(F_{n+2} - F_n).$$

As $F_{n+1} - F_{n-1} = F_n$ and $F_{n+2} - F_n = F_{n+1}$, this obviously holds. Therefore, we have

$$\begin{aligned} F_{n+1}^2 - F_n F_{n+2} &= -(F_n^2 - F_{n-1} F_{n+1}) = F_{n-1}^2 - F_{n-2} F_n \\ &= \dots = (-1)^n (F_1^2 - F_0 F_2) = (-1)^n. \end{aligned}$$

29. The relation can be rewritten as $a_n a_{n+2} - a_{n+1}^2 = 2$. Therefore,

$$a_n a_{n+2} - a_{n+1}^2 = a_{n-1} a_{n+1} - a_n^2$$

for any $n \geq 2$. This yields

$$a_n a_{n+2} + a_n^2 = a_{n-1} a_{n+1} + a_{n+1}^2$$

so that

$$\frac{a_{n+2} + a_n}{a_{n+1}} = \frac{a_{n+1} + a_{n-1}}{a_n}.$$

Thus, $\frac{a_{n+2} + a_n}{a_{n+1}}$ is a constant. In particular, we have

$$\frac{a_{n+2} + a_n}{a_{n+1}} = \frac{a_3 + a_1}{a_2} = 4.$$

This means $a_{n+2} = 4a_{n+1} - a_n$, which clearly shows that each a_n is an integer.

30. Define $a_2 = 1$, $a_3 = 3$ and $a_{n+2} = a_{n+1} + 2a_n$ for $n \geq 2$. From the small cases

$$u_2 = \frac{5}{2} = 2 + \frac{1}{2}, \quad u_3 = \frac{65}{8} = 8 + \frac{1}{8},$$

we guess $u_n = 2^{a_n} + 2^{-a_n}$ for $n \geq 2$.

Firstly, we observe that

$$a_{n+2} - 2a_{n+1} = -(a_{n+1} - 2a_n) = a_n - 2a_{n-1} = \dots = \pm(a_3 - 2a_2) = \pm 1.$$

Now, we prove the formula by induction on n . The base cases $n = 2, 3$ holds from the above values. Assume $u_n = 2^{a_n} + 2^{-a_n}$ holds for any $n = 2, 3, \dots, k$ where $k \geq 3$. Then

$$\begin{aligned}
u_{k+1} &= u_k(u_{k-1}^2 - 2) - u_1 \\
&= \left(2^{a_k} + \frac{1}{2^{a_k}}\right) \left(\left(2^{a_{k-1}} + \frac{1}{2^{a_{k-1}}}\right)^2 - 2\right) - \frac{5}{2} \\
&= \left(2^{a_k} + \frac{1}{2^{a_k}}\right) \left(2^{2a_{k-1}} + \frac{1}{2^{2a_{k-1}}} - 2\right) - \frac{5}{2} \\
&= 2^{a_k+2a_{k-1}} + \frac{1}{2^{a_k+2a_{k-1}}} + 2^{a_k-2a_{k-1}} + \frac{1}{2^{a_k-2a_{k-1}}} - \frac{5}{2} \\
&= 2^{a_{k+1}} + \frac{1}{2^{a_{k+1}}}
\end{aligned}$$

since $a_k - 2a_{k-1} = \pm 1$. This establishes the formula $u_n = 2^{a_n} + 2^{-a_n}$ for any $n \geq 2$ by induction.

Now, the characteristic equation of the sequence (a_n) is $\lambda^2 - \lambda - 2 = 0$. Its roots are 2, -1. Thus, $a_n = A \cdot 2^n + B \cdot (-1)^n$ for some constants A, B . Solving the system

$$\begin{cases} 4A + B = 1, \\ 8A - B = 3, \end{cases}$$

we obtain $A = \frac{1}{3}$ and $B = -\frac{1}{3}$. Hence,

$$u_n = 2^{\frac{2^n - (-1)^n}{3}} + \frac{1}{2^{\frac{2^n - (-1)^n}{3}}}$$

and its integer part is $2^{\frac{2^n - (-1)^n}{3}}$.

31. The characteristic equation of the sequence is $\lambda^2 - 7\lambda + 1 = 0$. Its roots are $\frac{7 \pm 3\sqrt{5}}{2}$. Thus,

$$a_n = A \left(\frac{7+3\sqrt{5}}{2}\right)^n + B \left(\frac{7-3\sqrt{5}}{2}\right)^n$$

for some constants A, B . Solving the system

$$\begin{cases} \left(\frac{7+3\sqrt{5}}{2}\right)A + \left(\frac{7-3\sqrt{5}}{2}\right)B = 1, \\ \left(\frac{7+3\sqrt{5}}{2}\right)^2 A + \left(\frac{7-3\sqrt{5}}{2}\right)^2 B = 1, \end{cases}$$

we obtain $A = \frac{9-4\sqrt{5}}{3}$ and $B = \frac{9+4\sqrt{5}}{3}$. Hence,

$$a_n = \frac{9-4\sqrt{5}}{3} \cdot \left(\frac{7+3\sqrt{5}}{2}\right)^n + \frac{9+4\sqrt{5}}{3} \cdot \left(\frac{7-3\sqrt{5}}{2}\right)^n.$$

From this, we have

$$\begin{aligned}
& a_n + a_{n+1} + 2 \\
&= \frac{9 - 4\sqrt{5}}{3} \cdot \left(\frac{7 + 3\sqrt{5}}{2}\right)^n \left(1 + \frac{7 + 3\sqrt{5}}{2}\right) + \frac{9 + 4\sqrt{5}}{3} \cdot \left(\frac{7 - 3\sqrt{5}}{2}\right)^n \left(1 + \frac{7 - 3\sqrt{5}}{2}\right) + 2 \\
&= \frac{7 - 3\sqrt{5}}{2} \cdot \left(\frac{7 + 3\sqrt{5}}{2}\right)^n + \frac{7 + 3\sqrt{5}}{2} \cdot \left(\frac{7 - 3\sqrt{5}}{2}\right)^n + 2 \\
&= \left(\frac{3 - \sqrt{5}}{2}\right)^2 \left(\frac{3 + \sqrt{5}}{2}\right)^{2n} + \left(\frac{3 + \sqrt{5}}{2}\right)^2 \left(\frac{3 - \sqrt{5}}{2}\right)^{2n} + 2 \\
&= \left(\left(\frac{3 - \sqrt{5}}{2}\right)\left(\frac{3 + \sqrt{5}}{2}\right)^n + \left(\frac{3 + \sqrt{5}}{2}\right)\left(\frac{3 - \sqrt{5}}{2}\right)^n\right)^2
\end{aligned}$$

since $\frac{3 + \sqrt{5}}{2} \cdot \frac{3 - \sqrt{5}}{2} = 1$. Let $a_n + a_{n+1} + 2 = b_n^2$. It remains to show that the expression b_n under the square is an integer. To see this, note that it is of the form $a + b\sqrt{5}$ for some rational numbers a, b . If we change every $\sqrt{5}$ to $-\sqrt{5}$, then b_n remains unchanged. This shows $a + b\sqrt{5} = a - b\sqrt{5}$ so that $b = 0$. Thus, b_n is rational. Since b_n^2 is an integer, b_n must be an integer. This completes the proof.

32. One may find the general formula and prove the assertion directly. Here, we present an alternative proof.

Firstly, we prove that $a_{n+1}^2 - a_n a_{n+2} = a_n^2 - a_{n-1} a_{n+1}$ for $n \geq 1$. Rearranging the terms, we need to prove

$$a_{n+1}(a_{n+1} + a_{n-1}) = a_n(a_{n+2} + a_n).$$

As $a_{n+1} + a_{n-1} = 3a_n$ and $a_{n+2} + a_n = 3a_{n+1}$, this obviously holds. Therefore, we have

$$a_{n+1}^2 - a_n a_{n+2} = a_n^2 - a_{n-1} a_{n+1} = \cdots = a_1^2 - a_0 a_2 = 1.$$

From $a_n + a_{n+2} = 3a_{n+1}$ and $a_n a_{n+2} = a_{n+1}^2 - 1$, the numbers a_n and a_{n+2} are the roots of

$$t^2 - 3a_{n+1}t + (a_{n+1}^2 - 1) = 0.$$

In particular, we have $a_n^2 - 3a_{n+1}a_n + (a_{n+1}^2 - 1) = 0$. It follows that

$$a_n a_{n+1} + 1 = a_n^2 - 2a_n a_{n+1} + a_{n+1}^2 = (a_n - a_{n+1})^2$$

is a perfect square.

33. Rewrite the equation as $(a_{n+1} - ka_n)^2 = (k^2 - 1)a_n^2 + 1$. This is the same as

$$a_{n+1}^2 - 2ka_n a_{n+1} + (a_n^2 - 1) = 0. \tag{1}$$

Replacing n by $n - 1$ and reordering the terms, we obtain

$$a_{n-1}^2 - 2ka_n a_{n-1} + (a_n^2 - 1) = 0. \tag{2}$$

It is obvious from the definition that $a_{n+1} > a_n > a_{n-1}$. Therefore, (1) and (2) show that a_{n+1} and a_{n-1} are the roots of the quadratic equation

$$t^2 - 2ka_nt + (a_n^2 - 1) = 0.$$

By Vieta's theorem, we have $a_{n-1} + a_{n+1} = 2ka_n$. As this holds for any $n \geq 1$, it is obvious by induction that each term is an integer. From this recurrence relation, we also find that $a_{n+1} \equiv -a_{n-1} \pmod{2k}$. As $2k \mid a_0$, we can show by induction that $2k \mid a_{2n}$ for every nonnegative integer n .

34. Note that $a_n > 0$ for each n . Thus, the formula is well-defined. Let

$$b_n = \sqrt{1 + 24a_n}$$

for $n \geq 1$. Then b_n is positive and the formula becomes

$$\frac{b_{n+1}^2 - 1}{24} = \frac{1}{16} \left(1 + 4 \cdot \frac{b_n^2 - 1}{24} + b_n \right).$$

This can be simplified to $4b_{n+1}^2 = b_n^2 + 6b_n + 9 = (b_n + 3)^2$. As b_n is positive, this implies

$$2b_{n+1} = b_n + 3.$$

This gives $2(b_{n+1} - 3) = b_n - 3$. Using this repeatedly, we have

$$b_n - 3 = \frac{1}{2}(b_{n-1} - 3) = \frac{1}{2^2}(b_{n-2} - 3) = \cdots = \frac{1}{2^{n-1}}(b_1 - 3) = \frac{1}{2^{n-2}}.$$

Therefore,

$$a_n = \frac{b_n^2 - 1}{24} = \frac{1}{24} \left(\left(\frac{1}{2^{n-2}} + 3 \right)^2 - 1 \right) = \frac{1}{3} \left(\frac{1}{2^{2n-1}} + \frac{3}{2^n} + 1 \right).$$